

Robust Controller Synthesis

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I. ODE IQC DUALITY

This section reviews the dual IQC formulation for robustness analysis in the ODE setting. The results of *Robust $\mathcal{L}_2$ -gain feedforward control of uncertain systems using dynamic IQCs* [3] and *Convex LPV synthesis of estimators and feedforwards using duality and integral quadratic constraints* [11] are the main finite-dimensional references. They will later serve as a point of comparison for the infinite-dimensional formulation based on PI operators and PIE systems.

A. Köse and Scherer

The starting point is the standard IQC robustness theorem.

Theorem 1 (Megretski and Rantzer [6]). *Suppose G is stable and*

- 1) *the feedback interconnection of $\tau\Delta$ and G is well-defined for all $\tau \in [0, 1]$;*
- 2) *$\tau\Delta$ satisfies the IQC defined by Π for all $\tau \in [0, 1]$, i.e.,*

$$\left\langle \begin{bmatrix} q \\ \tau\Delta q \end{bmatrix}, \Pi \begin{bmatrix} q \\ \tau\Delta q \end{bmatrix} \right\rangle \geq 0 \quad \forall q \in \mathcal{L}_{2+}, \quad \forall \tau \in [0, 1]; \quad (1)$$

- 3) *G satisfies*

$$\begin{bmatrix} G \\ I \end{bmatrix}^* \Pi \begin{bmatrix} G \\ I \end{bmatrix} \prec 0 \quad \text{on } \mathbb{C}_0. \quad (2)$$

Then the feedback interconnection of Δ and G is stable.

The dual IQC formulation used in [3] follows from a finite-dimensional inertia argument.

Lemma 1. *Let $S \in \mathbb{C}^{(m+n) \times m}$ have full column rank and let $M = M^* \in \mathbb{C}^{(m+n) \times (m+n)}$ satisfy $\text{in}(M) = (m, n, 0)$ (implying $M_{11}, -M_{22} > 0$). Then $S^*MS \prec 0$ if and only if $S_\perp^*MS_\perp \succ 0$, where $S_\perp$ forms a basis for the orthogonal complement of the image of S .*

Applying Lemma 1 to (2) gives the dual frequency-domain condition

$$\begin{bmatrix} G^* \\ I \end{bmatrix}^* \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \Pi^{-\top} \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \begin{bmatrix} G^* \\ I \end{bmatrix} \prec 0 \quad \text{on } \mathbb{C}_0. \quad (3)$$

For later use, define the dual multiplier

$$\mathbf{D}(\Pi) := \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \Pi^{-\top} \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} =: \Theta = \begin{bmatrix} \Theta_{11} & \Theta_{12} \\ \Theta_{12}^* & \Theta_{22} \end{bmatrix}. \quad (4)$$

Applying the same inertia argument to (1) yields the corresponding dual IQC

$$\left\langle \begin{bmatrix} q \\ \tau\Delta^*q \end{bmatrix}, \Theta \begin{bmatrix} q \\ \tau\Delta^*q \end{bmatrix} \right\rangle \leq 0 \quad \forall q \in \mathcal{L}_{2+}, \quad \forall \tau \in [0, 1]. \quad (5)$$

B. Venkataraman and Seiler

The duality framework in [11] uses this construction together with J -spectral factorizations of PN multipliers.

Definition 1 (Hu et al. [2]). *Let $\Pi = \Pi^\sim \in \mathbb{R}L_\infty^{(n_v+n_w) \times (n_v+n_w)}$, where $H^\sim(s) = H(-s)^\top$ for $H \in \mathbb{R}L_\infty$. Partition Π as*

$$\Pi = \begin{bmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{12}^\sim & \Pi_{22} \end{bmatrix},$$

where $\Pi_{11} \in \mathbb{R}L_\infty^{n_v \times n_v}$ and $\Pi_{22} \in \mathbb{R}L_\infty^{n_w \times n_w}$. Then Π is called a strict positive-negative (PN) multiplier if $\Pi_{11}(j\omega) > 0$ and $\Pi_{22}(j\omega) < 0$ for all $\omega \in \mathbb{R} \cup \{\infty\}$.

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Definition 2 (Bart et al. [1] and Meinsma [7]). *Let*

$$J_{n_v, n_w} := \begin{bmatrix} I_{n_v} & 0 \\ 0 & -I_{n_w} \end{bmatrix}.$$

A pair $(\hat{\Psi}, J_{n_v, n_w})$ is called a J -spectral factorization of $\Pi = \Pi^\sim$ if

$$\Pi = \hat{\Psi}^\sim J_{n_v, n_w} \hat{\Psi}, \quad \hat{\Psi}, \hat{\Psi}^{-1} \in \mathbb{RH}_\infty^{(n_v + n_w) \times (n_v + n_w)}. \quad (6)$$

Remark 1. The strict PN condition in Definition 1 is a convenient class of admissible multipliers, but it is stronger than what the duality argument itself requires. The essential requirement is the existence of a stable, bistable J -spectral factorization in the sense of Definition 2.

For example, an off-diagonal multiplier

$$\Pi = \begin{bmatrix} 0 & \Pi_{12} \\ \Pi_{12}^\sim & 0 \end{bmatrix}$$

is not strict PN, but it admits a J -spectral factorization whenever the factor below is stable and bistable:

$$\Pi = \hat{\Psi}^\sim J \hat{\Psi}, \quad \hat{\Psi} := \frac{1}{\sqrt{2}} \begin{bmatrix} I & \Pi_{12} \\ I & -\Pi_{12} \end{bmatrix}, \quad J := \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}. \quad (7)$$

More generally, consider

$$\Pi = \begin{bmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{12}^\sim & \Pi_{22} \end{bmatrix}, \quad \Pi_{22} < 0, \quad \Pi_{11} - \Pi_{12} \Pi_{22}^{-1} \Pi_{12}^\sim > 0.$$

Then Π admits the J -spectral factorization

$$\Pi = \Psi^\sim J \Psi, \quad \Psi := \begin{bmatrix} F_{11} & 0 \\ F_{22} \Pi_{22}^{-1} \Pi_{12}^\sim & F_{22} \end{bmatrix}, \quad J := \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}, \quad (8)$$

where

$$F_{11}^\sim F_{11} = \Pi_{11} - \Pi_{12} \Pi_{22}^{-1} \Pi_{12}^\sim, \quad F_{22}^\sim F_{22} = -\Pi_{22}.$$

This form includes the preceding examples after choosing the corresponding block factors.

The state-space version of the result in [11] can be described as follows. Suppose $\Pi = \hat{\Psi}^\sim J \hat{\Psi}$ is a J -spectral factorization as in Definition 2. Then the dual IQC can be written in factored form as

$$\left\langle \mathbf{D}(\hat{\Psi}) \begin{bmatrix} z_\Delta \\ \tau \Delta^* z_\Delta \end{bmatrix}, J \mathbf{D}(\hat{\Psi}) \begin{bmatrix} z_\Delta \\ \tau \Delta^* z_\Delta \end{bmatrix} \right\rangle \leq 0 \quad \forall z_\Delta \in \mathcal{L}_{2+}, \quad \forall \tau \in [0, 1]. \quad (9)$$

First write the generalized plant as

$$\begin{bmatrix} \dot{x}_G(t) \\ z_\Delta(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} A & B_\Delta & B_w \\ C_\Delta & D_\Delta & D_{\Delta w} \\ C_z & D_{z\Delta} & D_{zw} \end{bmatrix} \begin{bmatrix} x_G(t) \\ w_\Delta(t) \\ w(t) \end{bmatrix}, \quad (10)$$

where w and z denote the exogenous input and performance output, respectively. Next, let $\hat{\Psi} \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ have the realization

$$\begin{bmatrix} \dot{x}_\Psi(t) \\ \tilde{z}_\Delta(t) \\ \tilde{w}_\Delta(t) \end{bmatrix} = \begin{bmatrix} A_\Psi & B_\Psi^z & B_\Psi^w \\ C_\Psi^z & D_\Psi^{zz} & D_\Psi^{zw} \\ C_\Psi^w & D_\Psi^{wz} & D_\Psi^{ww} \end{bmatrix} \begin{bmatrix} x_\Psi(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}. \quad (11)$$

Let $x := \begin{bmatrix} x_G \\ x_\Psi \end{bmatrix}$. Augmenting the generalized plant (10) with the multiplier realization gives

$$\begin{bmatrix} \dot{x}(t) \\ \tilde{z}_\Delta(t) \\ \tilde{w}_\Delta(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B}_\Delta & \mathcal{B}_w \\ \mathcal{C}_z & \mathcal{D}_{z\Delta} & \mathcal{D}_{zw} \\ \mathcal{C}_{\tilde{w}} & \mathcal{D}_{\tilde{w}\Delta} & \mathcal{D}_{\tilde{w}w} \\ \mathcal{C}_z & \mathcal{D}_{z\Delta} & \mathcal{D}_{zw} \end{bmatrix} \begin{bmatrix} x(t) \\ w_\Delta(t) \\ w(t) \end{bmatrix}. \quad (12)$$

The calligraphic matrices in (12) are

$$\begin{aligned} \mathcal{A} &:= \begin{bmatrix} A & 0 \\ B_\Psi^z C_\Delta & A_\Psi \end{bmatrix}, & \mathcal{B}_\Delta &:= \begin{bmatrix} B_\Delta \\ B_\Psi^z D_\Delta + B_\Psi^w \end{bmatrix}, & \mathcal{B}_w &:= \begin{bmatrix} B_w \\ B_\Psi^z D_{\Delta w} \end{bmatrix}, \\ \mathcal{C}_z &:= \begin{bmatrix} D_\Psi^{zz} C_\Delta & C_\Psi^z \end{bmatrix}, & \mathcal{D}_{z\Delta} &:= D_\Psi^{zz} D_\Delta + D_\Psi^{zw}, & \mathcal{D}_{zw} &:= D_\Psi^{zz} D_{\Delta w}, \\ \mathcal{C}_{\tilde{w}} &:= \begin{bmatrix} D_\Psi^{wz} C_\Delta & C_\Psi^w \end{bmatrix}, & \mathcal{D}_{\tilde{w}\Delta} &:= D_\Psi^{wz} D_\Delta + D_\Psi^{ww}, & \mathcal{D}_{\tilde{w}w} &:= D_\Psi^{wz} D_{\Delta w}, \\ \mathcal{C}_z &:= \begin{bmatrix} C_z & 0 \end{bmatrix}, & \mathcal{D}_{z\Delta} &:= D_{z\Delta}, & \mathcal{D}_{zw} &:= D_{zw}. \end{aligned} \quad (13)$$

Figure 1 summarizes the primal-dual correspondence used in [11]. The passage from the top-right realization to the bottom-right realization can be read in the following steps.

- 1) Start with the primal interconnection in the top-right diagram, where the generalized plant is augmented with the J -spectral factor $\hat{\Psi}$ as in (12). This realization gives the primal dissipation inequality and the corresponding LMI.
- 2) One can then apply a congruence transformation on the LMI of the form

$$T = \begin{bmatrix} I & 0 & 0 \\ -\mathcal{D}_{\tilde{w}w}^{-1}\mathcal{C}_{\tilde{w}} & \mathcal{D}_{\tilde{w}w}^{-1} & -\frac{1}{\sqrt{\gamma}}\mathcal{D}_{\tilde{w}w}^{-1}\mathcal{D}_{\tilde{w}z} \\ 0 & 0 & \frac{1}{\sqrt{\gamma}}I \end{bmatrix}. \quad (14)$$

This gives the system on the top-left diagram and shows equivalent LMI conditions. The factor γ is included in [11], but is not essential to the duality argument itself. Avoiding this scaling leaves the framework open to other performance descriptions, such as passivity, H_∞ performance, and quadratic performance.

- 3) The top-left realization is the same augmented plant-multiplier interconnection after the algebraic loop has been closed by this substitution. In realization coordinates, this can be interpreted as a variable reorganization of the spectral factor, but the LMI step only uses the congruence.
- 4) Apply Lemma 1 to the top-left primal system. This produces the dual realization shown in the bottom-left diagram.
- 5) After applying the inverse congruence on the dual side, the bottom-right system is LMI-equivalent to the bottom-left system. In this final form the scaled signals represent an input-output mapping, which is the representation used for synthesis in [11].

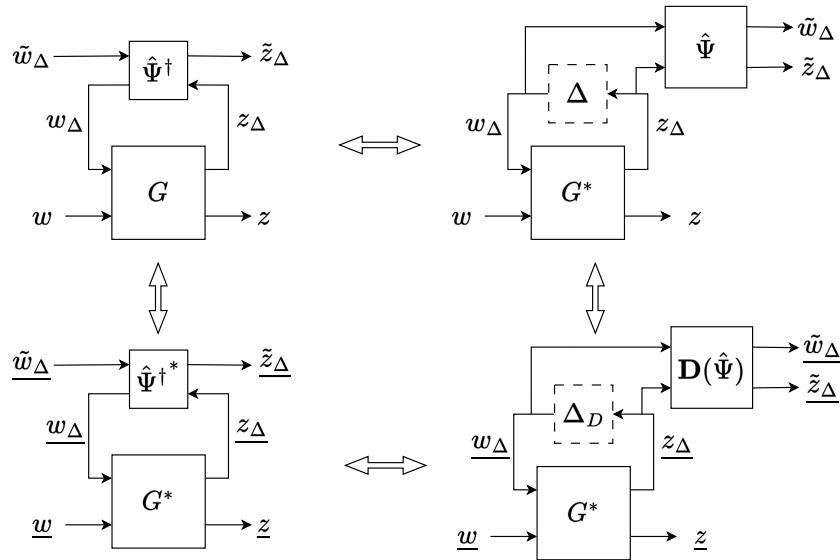


Fig. 1. Primal-dual correspondence in [11]. Venkataraman and Seiler include an α -scaling of the multiplier. In their notation $\alpha = \sqrt{\gamma}$; here this scaling is avoided by using a different congruence transformation.

II. PIE IQC DUALITY

This section gives the PIE analogue of the finite-dimensional overview in the preceding section. The central point is that the primal-dual construction can be written directly in terms of PI operators: the primal scaling defines a scaled input-output map, the adjoint PIE defines the dual dynamics, and the dual scaling is determined algebraically from the primal scaling.

Definition 3. For $w(t) \in \mathbb{R}L_2^{n_w}$, $\mathbf{x}(t) \in \mathbb{R}L_2^{n_x}$, and $z(t) \in \mathbb{R}L_2^{n_z}$ that satisfy $\mathbf{x}(0) = 0$, the **Primal System** $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ is defined as,

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \quad (15)$$

where $\mathcal{T}, \mathcal{A} \in \Pi_4^{n_x \times n_x}$, $\mathcal{B} \in \Pi_4^{n_w \times n_x}$, $\mathcal{C} \in \Pi_4^{n_x \times n_z}$, and $\mathcal{D} \in \Pi_4^{n_w \times n_z}$ are PI operators. The **Scaled Primal System** is expressed in terms of $\tilde{w}(t) \in \mathbb{R}L_2^{n_w}$ and $\tilde{z}(t) \in \mathbb{R}L_2^{n_z}$, which satisfy,

$$\begin{bmatrix} z(t) \\ w(t) \end{bmatrix} = \underbrace{\begin{bmatrix} \mathcal{S}_{11} & \mathcal{S}_{12} \\ \mathcal{S}_{21} & \mathcal{S}_{22} \end{bmatrix}}_S \begin{bmatrix} \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix}. \quad (16)$$

The operator $\mathcal{S}$ is called the **Primal I/O Scaling** of the system.

Definition 4. For $\underline{w}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$, $\underline{x}(t) \in \mathbb{R}L_2^{\mathbf{n}_x}$, and $\underline{z}(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$ that satisfy $\underline{x}(0) = 0$, the **Dual System** $G\{\underline{\mathcal{T}}, \underline{\mathcal{A}}, \underline{\mathcal{B}}, \underline{\mathcal{C}}, \underline{\mathcal{D}}\} = G\{\mathcal{T}^*, \mathcal{A}^*, \mathcal{C}^*, \mathcal{B}^*, \mathcal{D}^*\}$ is defined as,

$$\begin{bmatrix} \mathcal{T}^* \dot{\underline{x}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{x}(t) \\ \underline{w}(t) \end{bmatrix}. \quad (17)$$

The **Scaled Dual System** is expressed in terms of $\underline{\tilde{w}}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$ and $\underline{\tilde{z}}(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$, which satisfy,

$$\begin{bmatrix} \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} = \underbrace{\begin{bmatrix} \mathcal{S}_{22}^* & -\mathcal{S}_{12}^* \\ -\mathcal{S}_{21}^* & \mathcal{S}_{11}^* \end{bmatrix}^{-1}}_{\underline{\mathcal{S}}} \begin{bmatrix} \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix}. \quad (18)$$

The operator $\underline{\mathcal{S}}$ is called the **Dual I/O Scaling** of the system and is parameterized by entries from the **Primal I/O Scaling**.

Theorem 2 (Scaled IO duality). Assume the following conditions hold

- The **Primal I/O Scaling** is a PI operator and invertible.
- $\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21}$ and $\underline{\mathcal{S}}_{11} - \mathcal{D}^*\underline{\mathcal{S}}_{21}$ are invertible.

The following statements are equivalent.

- 1) For all $\tilde{w} \in \mathbb{R}L_2^{\mathbf{n}_w}$, the **Scaled Primal System** satisfies $\|P_\tau \tilde{z}\| \leq \|P_\tau \tilde{w}\|$.
- 2) For all $\tilde{w} \in \mathbb{R}L_2^{\mathbf{n}_z}$, the **Scaled Dual System** satisfies $\|P_\tau \tilde{z}\| \leq \|P_\tau \tilde{w}\|$.

The PIE generalized plant used in the IQC construction is

$$\begin{aligned} \mathcal{T} \dot{\mathbf{x}}_f(t) &= \mathcal{A} \mathbf{x}_f(t) + \mathcal{B}_\Delta w_\Delta(t) + \mathcal{B}_w w(t), \\ z_\Delta(t) &= \mathcal{C}_\Delta \mathbf{x}_f(t) + \mathcal{D}_\Delta w_\Delta(t) + \mathcal{D}_{\Delta w} w(t), \\ z(t) &= \mathcal{C}_z \mathbf{x}_f(t) + \mathcal{D}_{z\Delta} w_\Delta(t) + \mathcal{D}_{zw} w(t). \end{aligned} \quad (19)$$

The uncertainty IQC is imposed on the filtered signal

$$v(t) = \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}, \quad \langle p_T v, p_T \mathcal{M}_{\mathcal{V}_\Delta} v \rangle_{\mathbf{L}_{[a,b]}} \geq 0. \quad (20)$$

The filter Ψ is often tall rather than square [3], [10]. To obtain an invertible coordinate transform, we complete it to a square filter. More generally, the completion is described by a projection from the square coordinates back to the original IQC coordinates and by a right inverse of that projection. Thus we choose operators $\mathcal{N}_v$ and $\mathcal{E}_v$ satisfying

$$v = \mathcal{N}_v v_\square, \quad \mathcal{N}_v \mathcal{E}_v = I. \quad (21)$$

The special choice $\mathcal{E} = \mathcal{N}^*$ corresponds to orthogonal zero-completion, but the construction does not rely on that choice. The IQC in the square coordinates is defined using a freely chosen square-coordinate multiplier $\mathcal{M}_{\mathcal{V}_\Delta}^\square$:

$$\langle p_T v_\square, p_T \mathcal{M}_{\mathcal{V}_\Delta} v_\square \rangle_{\mathbf{L}_{[a,b]}} \geq 0, \quad (22)$$

where $\mathcal{M}_{\mathcal{V}_\Delta}$ is selected directly in the completed coordinates.

What needs to be done before applying Theorem 2 in case we want to use dynamic multipliers,

- 1) Start with the generalized PIE plant (19) and the dynamic IQC filter Ψ in (20). At this point the uncertainty is described in the original filtered coordinates v .
- 2) Square up the filter. The projection $\mathcal{N}$ records which part of the square signal $v_\square$ enters the original IQC, while $\mathcal{E}$ records how original coordinates are embedded into the square coordinates.
- 3) Augment the generalized plant with the squared-up filter realization. If the uncertainty channel itself is also completed, use the same idea: project square inputs back to the original channel with a map $\mathcal{N}$ and embed original outputs with a right inverse $\mathcal{E}$ satisfying $\mathcal{N}\mathcal{E} = I$. The corresponding signal route is

$$w_{\Delta,\square} \xrightarrow{\mathcal{N}} w_\Delta \xrightarrow{G} z_\Delta \xrightarrow{\mathcal{E}} z_{\Delta,\square} \xrightarrow{\Psi_\square} v_\square.$$

If $\mathcal{N}$ or $\mathcal{E}$ is dynamic, its state is included in the augmented realization; the algebraic loop only sees the corresponding feedthrough operators.

- 4) Close the resulting algebraic loop by the corresponding congruence transformation. This is the coordinate-free way of obtaining the scaled input-output map; a realization-level variable exchange is only one interpretation of the same substitution.
- 5) Factor the chosen static PI multiplier as $\mathcal{V}_\Delta = (\mathcal{S})^* J \mathcal{S}$, when such an invertible factorization is available. The congruence-closed realization gives the scaled primal input-output map used in Theorem 2.

Apply Theorem 2. The 'adjoint' PIE gives the dual dynamics, and (18) gives the dual scaling $\underline{\mathcal{S}}$ from the primal scaling $\mathcal{S}$. The resulting lower-right system is the scaled dual input-output map used for synthesis after constructing the dual multiplier $\mathbf{D}(\mathcal{V}) = \underline{\mathcal{S}}^{-*} J \underline{\mathcal{S}}^{-1}$, with the same bound as the scaled primal system. Figure 2 summarizes the PIE duality construction according to Theorem 2.

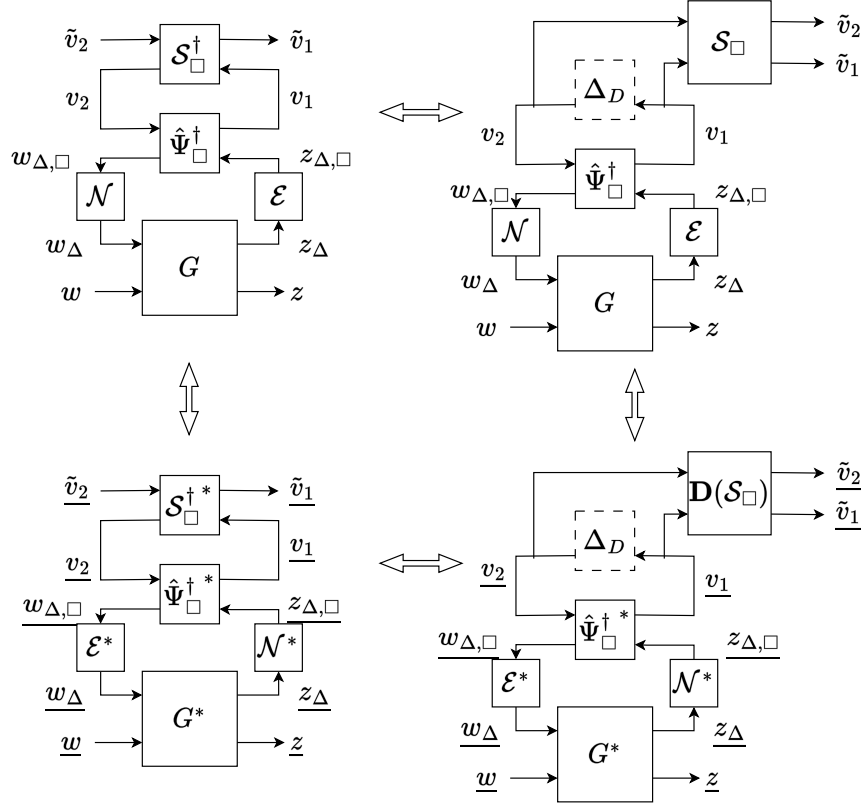


Fig. 2. Primal-dual correspondence for scaled PIE systems, adapted from [4]. The primal scaling induces the dual scaling, so both scaled systems represent equivalent input-output gain conditions.

III. LPI DERIVATION

Start with the squared-up dynamic multiplier $\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} = \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}$,

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ v_1(t) \\ v_2(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi & \mathcal{B}_\Psi^z & \mathcal{B}_\Psi^w \\ \mathcal{C}_\Psi^z & \mathcal{D}_\Psi^z & \mathcal{D}_\Psi^{zw} \\ \mathcal{C}_\Psi^w & \mathcal{D}_\Psi^w & \mathcal{D}_\Psi^{ww} \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}. \quad (23)$$

To square up the interconnection, introduce a full-row-rank map $\mathcal{N}$ from the squared coordinates to the original coordinates, and choose any right inverse $\mathcal{E}$ satisfying

$$\mathcal{N}\mathcal{E} = I. \quad (24)$$

The choice $\mathcal{E} = \mathcal{N}^*$ is the usual zero-padding choice when $\mathcal{N}$ is an orthogonal selector, but nothing below requires this. The original plant sees the projected input $\mathcal{N}w_\Delta$, while its original uncertainty output is embedded into the squared coordinates by $\mathcal{E}$. The signal route is

$$w_{\Delta,\square} \xrightarrow{\mathcal{N}} w_{\Delta,o} \xrightarrow{G} z_{\Delta,o} \xrightarrow{\mathcal{E}} z_{\Delta,\square} \xrightarrow{\Psi_\square^\dagger} w_{\Delta,\square}. \quad (25)$$

The maps $\mathcal{N}$ and $\mathcal{E}$ appear in the plant blocks. If these maps are dynamic, their states are appended to the augmented realization; the algebraic loop below is then obtained from their direct feedthrough operators. Thus the squared-up generalized plant is

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}} \\ z_{\Delta,\square} \\ z \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B}_\Delta \mathcal{N} & \mathcal{B}_w \\ \mathcal{E}\mathcal{C}_\Delta & \mathcal{E}\mathcal{D}_\Delta \mathcal{N} & \mathcal{E}\mathcal{D}_{\Delta w} \\ \mathcal{C}_z & \mathcal{D}_{z\Delta} \mathcal{N} & \mathcal{D}_{zw} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ w_{\Delta,\square} \\ w \end{bmatrix}. \quad (26)$$

Use the abbreviations

$$\begin{aligned} \mathcal{C}_\Delta^\square &:= \mathcal{E}\mathcal{C}_\Delta, & \mathcal{D}_\Delta^\square &:= \mathcal{E}\mathcal{D}_\Delta\mathcal{N}, & \mathcal{D}_{\Delta w}^\square &:= \mathcal{E}\mathcal{D}_{\Delta w}, \\ \mathcal{B}_\Delta^\square &:= \mathcal{B}_\Delta\mathcal{N}, & \mathcal{D}_{z\Delta}^\square &:= \mathcal{D}_{z\Delta}\mathcal{N}, & \mathbf{x}^a &:= \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_\Psi \end{bmatrix}, & \mathcal{T}^a &:= \begin{bmatrix} \mathcal{T} & 0 \\ 0 & \mathcal{T}_\Psi \end{bmatrix}. \end{aligned} \quad (27)$$

Substituting the squared-up plant signals into the multiplier realization and stacking the plant and multiplier states gives the open augmented realization with state $\mathbf{x}^a = \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_\Psi \end{bmatrix}$. Before closing the algebraic loop, this gives

$$\begin{bmatrix} \mathcal{T}^a \dot{\mathbf{x}}^a(t) \\ v_1(t) \\ v_2(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^a & \mathcal{B}_\Delta^a & \mathcal{B}_w^a \\ \mathcal{C}_1^a & \mathcal{D}_{11}^a & \mathcal{D}_{12}^a \\ \mathcal{C}_2^a & \mathcal{D}_{21}^a & \mathcal{D}_{22}^a \\ \mathcal{C}_z^a & \mathcal{D}_{z\Delta}^\square & \mathcal{D}_{zw} \end{bmatrix} \begin{bmatrix} \mathbf{x}^a(t) \\ w_{\Delta,\square}(t) \\ w(t) \end{bmatrix}, \quad (28)$$

where

$$\begin{aligned} \mathcal{A}^a &:= \begin{bmatrix} \mathcal{A} & 0 \\ \mathcal{B}_\Psi^z \mathcal{C}_\Delta^\square & \mathcal{A}_\Psi \end{bmatrix}, & \mathcal{B}_\Delta^a &:= \begin{bmatrix} \mathcal{B}_\Delta^\square \\ \mathcal{B}_\Psi^z \mathcal{D}_\Delta^\square + \mathcal{B}_\Psi^w \end{bmatrix}, & \mathcal{B}_w^a &:= \begin{bmatrix} \mathcal{B}_w \\ \mathcal{B}_\Psi^z \mathcal{D}_{\Delta w}^\square \end{bmatrix}, \\ \mathcal{C}_1^a &:= [\mathcal{D}_\Psi^z \mathcal{C}_\Delta^\square \quad \mathcal{C}_\Psi^z], & \mathcal{D}_{11}^a &:= \mathcal{D}_\Psi^z \mathcal{D}_\Delta^\square + \mathcal{D}_\Psi^w, & \mathcal{D}_{12}^a &:= \mathcal{D}_\Psi^z \mathcal{D}_{\Delta w}^\square, \\ \mathcal{C}_2^a &:= [\mathcal{D}_\Psi^{wz} \mathcal{C}_\Delta^\square \quad \mathcal{C}_\Psi^w], & \mathcal{D}_{21}^a &:= \mathcal{D}_\Psi^{wz} \mathcal{D}_\Delta^\square + \mathcal{D}_\Psi^w, & \mathcal{D}_{22}^a &:= \mathcal{D}_\Psi^{wz} \mathcal{D}_{\Delta w}^\square, \\ \mathcal{C}_z^a &:= [\mathcal{C}_z \quad 0]. \end{aligned} \quad (29)$$

The algebraic loop is entirely contained in the v_2 -row:

$$v_2 = \mathcal{C}_2^a \mathbf{x}^a + \mathcal{D}_{21}^a w_{\Delta,\square} + \mathcal{D}_{22}^a w. \quad (30)$$

Assume that $\mathcal{D}_{21}^a$ is invertible. Then

$$w_{\Delta,\square} = -(\mathcal{D}_{21}^a)^{-1} \mathcal{C}_2^a \mathbf{x}^a + (\mathcal{D}_{21}^a)^{-1} v_2 - (\mathcal{D}_{21}^a)^{-1} \mathcal{D}_{22}^a w. \quad (31)$$

Hence the change of variables

$$\begin{bmatrix} \mathbf{x}^a \\ w_{\Delta,\square} \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} I & 0 & 0 \\ -(\mathcal{D}_{21}^a)^{-1} \mathcal{C}_2^a & (\mathcal{D}_{21}^a)^{-1} & -(\mathcal{D}_{21}^a)^{-1} \mathcal{D}_{22}^a \\ 0 & 0 & I \end{bmatrix}}_{\mathcal{X}_\Delta} \begin{bmatrix} \mathbf{x}^a \\ v_2 \\ w \end{bmatrix} \quad (32)$$

closes the loop. This is the congruence transformation: any quadratic form L in the variables $\begin{bmatrix} \mathbf{x}^a \\ w_{\Delta,\square} \\ w \end{bmatrix}$ becomes

$$L \mapsto \mathcal{X}_\Delta^* L \mathcal{X}_\Delta \quad (33)$$

in the variables $\begin{bmatrix} \mathbf{x}^a \\ v_2 \\ w \end{bmatrix}$. The inverse change of variables is just the original loop row (30):

$$\begin{bmatrix} \mathbf{x}^a \\ v_2 \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} I & 0 & 0 \\ \mathcal{C}_2^a & \mathcal{D}_{21}^a & \mathcal{D}_{22}^a \\ 0 & 0 & I \end{bmatrix}}_{\mathcal{X}_\Delta^{-1}} \begin{bmatrix} \mathbf{x}^a \\ w_{\Delta,\square} \\ w \end{bmatrix}. \quad (34)$$

Example 1 (Zero-padding completion). *For the trivial zero-padding completion the well-posedness implication is visible block by block. Let*

$$\mathcal{N} = \begin{bmatrix} I & 0 \end{bmatrix}, \quad \mathcal{E} = \mathcal{N}^* = \begin{bmatrix} I \\ 0 \end{bmatrix}, \quad w_{\Delta,\square} = \begin{bmatrix} w_{\Delta,o} \\ w_{\Delta,p} \end{bmatrix}, \quad (35)$$

so that $\mathcal{N} w_{\Delta,\square} = w_{\Delta,o}$, returns the original uncertainty input. Suppose $(I + \mathcal{L})$ denotes the well-posedness condition. $\mathcal{L}$ can be separated in a part $\mathcal{F}$ and $\mathcal{N}$. Since $\mathcal{F}$ can be separated in parts acting on the original coordinates and the padded coordinates, write

$$\mathcal{F} = \begin{bmatrix} \mathcal{F}_o \\ \mathcal{F}_p \end{bmatrix}. \quad (36)$$

Then the squared-up, and non-squared-up algebraic loop operators are

$$I + \mathcal{F}\mathcal{N} = \begin{bmatrix} I + \mathcal{F}_o & 0 \\ \mathcal{F}_p & I \end{bmatrix}, \quad I + \mathcal{N}\mathcal{F} = I + \mathcal{F}_o, \quad (37)$$

respectively. Thus the squared-up loop is block lower triangular: the padded directions add only an identity block, and the only possible singularity is the original non-squared-up loop $I + \mathcal{F}_o$.

Definition 5. Let $\mathcal{V} = \mathcal{V}^* \in \Pi_4$ be partitioned as

$$\mathcal{V} = \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix}.$$

Then $\mathcal{V}$ is called a strict PN-PI operator if there exists $\epsilon > 0$ such that

$$\mathcal{Q} \succeq \epsilon I, \quad \mathcal{R} \preceq -\epsilon I.$$

Lemma 2. Let $\mathcal{V} = \mathcal{V}^* \in \Pi_4$ and $\bar{\mathcal{V}} = \bar{\mathcal{V}}^* \in \Pi_4$ denote the static primal and dual IQC multipliers. Then $\mathcal{V}$ is a strict PN-PI operator if and only if $\bar{\mathcal{V}}$ is a strict PN-PI operator.

Proof. See Appendix III-B. ■

Example 2 (Dual multiplier for the zero-padded LTI-parametric IQC). In this example we assume that the scaled duality theorem holds for dynamic scalings with invertible feedthrough, and we use the square-up construction to apply it to the non-invertible factor. Consider the case of LTI parametric uncertainty. The primal IQC multiplier is given by

$$\int_0^T \left\langle \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}, \begin{bmatrix} \Psi_v^* & 0 \\ 0 & \Psi_v^* \end{bmatrix} \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix} \begin{bmatrix} \Psi_v & 0 \\ 0 & \Psi_v \end{bmatrix} \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix} \right\rangle dt \geq 0, \quad (38)$$

where Ψ_v is stable, $\mathcal{Q} \succ 0$, $\mathcal{R} \prec 0$, and $\mathcal{S} = -\mathcal{S}^*$. Usually Ψ_v is tall, with $\Psi_v = \begin{bmatrix} I \\ \Psi_N \end{bmatrix}$, where N denotes the spectral order of the dynamic multiplier. Complete this factor by

$$\Psi_{v,\square} = \begin{bmatrix} I & -\Psi_N^* \\ \Psi_N & I \end{bmatrix}, \quad \Psi_\square = \begin{bmatrix} \Psi_{v,\square} & 0 \\ 0 & \Psi_{v,\square} \end{bmatrix}. \quad (39)$$

The squared up primal IQC then becomes

$$\int_0^T \left\langle \begin{bmatrix} z_{\Delta,\square}(t) \\ w_{\Delta,\square}(t) \end{bmatrix}, \begin{bmatrix} \Psi_{v,\square}^* & 0 \\ 0 & \Psi_{v,\square}^* \end{bmatrix} \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix} \begin{bmatrix} \Psi_{v,\square} & 0 \\ 0 & \Psi_{v,\square} \end{bmatrix} \begin{bmatrix} z_{\Delta,\square}(t) \\ w_{\Delta,\square}(t) \end{bmatrix} \right\rangle dt \geq 0, \quad (40)$$

with $z_{\Delta,\square} = \begin{bmatrix} z_\Delta^\top & 0^\top \end{bmatrix}^\top$ and $w_{\Delta,\square} = \begin{bmatrix} w_\Delta^\top & 0^\top \end{bmatrix}^\top$ denoting the zero-padded signals. The multiplication operator $\mathcal{M}$ admits a J -factorization

$$\mathcal{M} = \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix} = \mathcal{V}^* \mathcal{J} \mathcal{V}, \quad \mathcal{V} := \begin{bmatrix} \mathcal{V}_{11} & \mathcal{V}_{12} \\ 0 & \mathcal{V}_{22} \end{bmatrix}, \quad \mathcal{J} := \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}, \quad (41)$$

where $\mathcal{S}_{11}$ and $\mathcal{S}_{22}$ are invertible. Hence the squared IQC multiplier is

$$\Pi_\square = \Psi_\square^* \mathcal{M} \Psi_\square = (\mathcal{V} \Psi_\square)^* \mathcal{J} (\mathcal{V} \Psi_\square). \quad (42)$$

The corresponding square primal scaling is the J -factor applied after the square dynamic filter:

$$\mathcal{L} := \mathcal{V} \Psi_\square = \begin{bmatrix} \mathcal{S}_{11} \Psi_{v,\square} & \mathcal{S}_{12} \Psi_{v,\square} \\ 0 & \mathcal{S}_{22} \Psi_{v,\square} \end{bmatrix}. \quad (43)$$

Theorem 2 uses the inverse signal route. Thus define

$$\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} = \mathcal{L}^{-1} \begin{bmatrix} \tilde{z}_\Delta \\ \tilde{w}_\Delta \end{bmatrix}. \quad (44)$$

If

$$\mathcal{A}_L := \mathcal{S}_{11} \Psi_{v,\square}, \quad \mathcal{B}_L := \mathcal{S}_{12} \Psi_{v,\square}, \quad \mathcal{D}_L := \mathcal{S}_{22} \Psi_{v,\square}, \quad (45)$$

then

$$\mathcal{L}^{-1} = \begin{bmatrix} \mathcal{A}_L^{-1} & -\mathcal{A}_L^{-1} \mathcal{B}_L \mathcal{D}_L^{-1} \\ 0 & \mathcal{D}_L^{-1} \end{bmatrix}. \quad (46)$$

Applying (18) gives the dual I/O scaling

$$\underline{\mathcal{L}}^{-1} = \begin{bmatrix} \mathcal{D}_L^{-*} & \mathcal{D}_L^{-*} \mathcal{B}_L^* \mathcal{A}_L^{-*} \\ 0 & \mathcal{A}_L^{-*} \end{bmatrix}^{-1}. \quad (47)$$

Therefore

$$\underline{\mathcal{L}} = \underbrace{\begin{bmatrix} \mathcal{S}_{22}^{-*} & \mathcal{S}_{22}^{-*} \mathcal{S}_{12}^* \mathcal{S}_{11}^{-*} \\ 0 & \mathcal{S}_{11}^{-*} \end{bmatrix}}_{\underline{\mathcal{V}}} \underbrace{\begin{bmatrix} \Psi_{v,\square}^{-*} & 0 \\ 0 & \Psi_{v,\square}^{-*} \end{bmatrix}}_{\underline{\Psi}_{\square}}. \quad (48)$$

The corresponding square dual multiplier is

$$\underline{\Pi}_{\square} = \underline{\mathcal{L}}^* J \underline{\mathcal{L}} = \underline{\Psi}_{\square}^* \underbrace{\underline{\mathcal{V}}^* J \underline{\mathcal{V}}}_{\underline{\mathcal{M}}} \underline{\Psi}_{\square}. \quad (49)$$

Equivalently, use the same static dual construction as in Lemma 2. If

$$\mathcal{M} = \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix}, \quad \mathcal{K} := \begin{bmatrix} -\mathcal{R} & \mathcal{S}^* \\ \mathcal{S} & -\mathcal{Q} \end{bmatrix}, \quad (50)$$

then the square static dual multiplier is

$$\underline{\mathcal{M}} := \mathcal{K}^{-1} = \begin{bmatrix} \underline{\mathcal{Q}} & \underline{\mathcal{S}} \\ \underline{\mathcal{S}}^* & \underline{\mathcal{R}} \end{bmatrix}. \quad (51)$$

Applying the block inverse formula with the Schur complement

$$\Theta := \mathcal{S}^* \mathcal{Q}^{-1} \mathcal{S} - \mathcal{R} \quad (52)$$

gives

$$\underline{\mathcal{Q}} = \Theta^{-1} \succ 0, \quad \underline{\mathcal{S}} = \Theta^{-1} \mathcal{S}^* \mathcal{Q}^{-1}, \quad (53)$$

$$\underline{\mathcal{R}} = -(\mathcal{Q} + \mathcal{S}(-\mathcal{R})^{-1} \mathcal{S}^*)^{-1} \prec 0. \quad (54)$$

Thus the square dual IQC condition is

$$\int_0^T \left\langle \begin{bmatrix} z_{\Delta}(t) \\ w_{\Delta}(t) \end{bmatrix}, \underline{\Pi}_{\square} \begin{bmatrix} z_{\Delta}(t) \\ w_{\Delta}(t) \end{bmatrix} \right\rangle dt \leq 0. \quad (55)$$

To unsquare this condition, let

$$\mathcal{N} = \begin{bmatrix} I & 0 \end{bmatrix}, \quad \mathcal{N}^* = \begin{bmatrix} I \\ 0 \end{bmatrix}. \quad (56)$$

The adjoint square-up embeds the non-squared dual uncertainty signals as

$$\begin{bmatrix} z_{\Delta,\square} \\ w_{\Delta,\square} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathcal{N}^* & 0 \\ 0 & \mathcal{N}^* \end{bmatrix}}_{\underline{\mathcal{E}}} \begin{bmatrix} z_{\Delta,o} \\ w_{\Delta,o} \end{bmatrix}. \quad (57)$$

Hence the non-squared dual multiplier is

$$\underline{\Pi}_o = \underline{\mathcal{E}}^* \underline{\Pi}_{\square} \underline{\mathcal{E}} = (\underline{\mathcal{L}} \underline{\mathcal{E}})^* J (\underline{\mathcal{L}} \underline{\mathcal{E}}). \quad (58)$$

For the zero-padding completion used here,

$$\underline{\Psi}_v = \Psi_{v,\square}^{-*} \mathcal{N}^* = \begin{bmatrix} (I + \Psi_N^* \Psi_N)^{-1} & -(I + \Psi_N^* \Psi_N)^{-1} \Psi_N^* \\ \Psi_N (I + \Psi_N^* \Psi_N)^{-1} & I - \Psi_N (I + \Psi_N^* \Psi_N)^{-1} \Psi_N^* \end{bmatrix} \begin{bmatrix} I \\ 0 \end{bmatrix} = (I + \Psi_N^* \Psi_N)^{-1} \Psi_v. \quad (59)$$

The unsquared dual IQC is then

$$\int_0^T \left\langle \begin{bmatrix} z_{\Delta}(t) \\ w_{\Delta}(t) \end{bmatrix}, \begin{bmatrix} \underline{\Psi}_v^* & 0 \\ 0 & \underline{\Psi}_v^* \end{bmatrix} \begin{bmatrix} \underline{\mathcal{Q}} & \underline{\mathcal{S}} \\ \underline{\mathcal{S}}^* & \underline{\mathcal{R}} \end{bmatrix} \begin{bmatrix} \underline{\Psi}_v & 0 \\ 0 & \underline{\Psi}_v \end{bmatrix} \begin{bmatrix} z_{\Delta}(t) \\ w_{\Delta}(t) \end{bmatrix} \right\rangle dt \leq 0. \quad (60)$$

which is also in the class of LTI parametric IQCs. Since all operations are invertible, we can start with the dual system and end up with a multiplier that satisfies the primal IQC. A convenient choice of the dynamic multiplier is all-pass basis function (12B) in [10]

$$\Psi_v = \begin{bmatrix} I & \frac{s+\rho}{s-\rho} & \frac{(s+\rho)^2}{(s-\rho)^2} & \dots & \frac{(s+\rho)^v}{(s-\rho)^v} \end{bmatrix}^T, \quad (61)$$

with $\rho < 0$. This choice is convenient as $\Psi_N^* \Psi_N$ is a scalar. Indeed,

$$\Psi_N^* \Psi_N = \sum_{i=1}^v \left(\frac{s-\rho}{s+\rho} \right)^i \left(\frac{s+\rho}{s-\rho} \right)^i = v \quad (62)$$

A. Proof of Theorem 2

Sublemma 1. *The **Primal I/O Scaling** is a PI operator and invertible if and only if the **Dual I/O Scaling** is a PI operator and invertible*

Proof. To show sufficiency, let $\bar{\mathcal{S}} = (\hat{\mathcal{J}}^* \mathcal{S} \hat{\mathcal{J}})^{-*} = \hat{\mathcal{J}}^* \mathcal{S}^{-*} \hat{\mathcal{J}} = \hat{\mathcal{J}}^* (\mathcal{S}^{-1})^* \hat{\mathcal{J}}$, where $\hat{\mathcal{J}} = \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix} \in \Pi_4$. Then if $\mathcal{S}$ is invertible then $\mathcal{S}^{-1} \in \Pi_4$ and by using that Π_4 forms a composition algebra we find that the **Dual I/O Scaling** is PI and invertible. For the necessary case, notice $\mathcal{S} = (\hat{\mathcal{J}}^* \bar{\mathcal{S}} \hat{\mathcal{J}})^{-*}$. ■

Sublemma 2. *If $\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21}$ and $\bar{\mathcal{S}}_{11} - \mathcal{D}^* \bar{\mathcal{S}}_{21}$ are invertible, then the **Primal Scaled System** and **Dual Scaled System** are well-posed, respectively.*

Proof. Applying the **Primal I/O Scaling** on the **Primal System** yields,

$$\begin{aligned} \partial_t(\mathcal{T}\mathbf{x})(t) &= \mathcal{A}\mathbf{x}(t) + \mathcal{B}(\mathcal{S}_{21}\tilde{z}(t) + \mathcal{S}_{22}\tilde{w}(t)), \\ (\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21})\tilde{z}(t) &= \mathcal{C}\mathbf{x}(t) + (\mathcal{D}\mathcal{S}_{22} - \mathcal{S}_{12})\tilde{w}(t). \end{aligned}$$

Then **Primal Scaled System** well-posed if and only if $\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21}$ is invertible, ensuring a unique solution for $\tilde{z}(t)$ in terms of $\mathbf{x}(t)$ and $\tilde{w}(t)$. A similar proof show that **Dual Scaled System** is well-posed if $\bar{\mathcal{S}}_{11} - \mathcal{D}^* \bar{\mathcal{S}}_{21}$ is invertible. ■

To show sufficiency, let $\mathbf{x}(t)$, $w(t)$, and $z(t)$ satisfy the primal system dynamics with $\mathbf{x}(0) = 0$, and let $\tilde{z}, \tilde{w}$ be defined by the primal scaling in 1):

$$z = \mathcal{S}_{11}\tilde{z} + \mathcal{S}_{12}\tilde{w}, \quad w = \mathcal{S}_{21}\tilde{z} + \mathcal{S}_{22}\tilde{w}. \quad (63)$$

From the proof of Theorem 9 in [9], specifically (12), we established that the unscaled signals, $z(t)$, $w(t)$, $\bar{z}(t)$ and $\bar{w}(t)$, satisfy the identity:

$$\int_0^t \langle \bar{z}(\theta), w(t-\theta) \rangle d\theta = \int_0^t \langle \bar{w}(\theta), z(t-\theta) \rangle d\theta. \quad (64)$$

Substituting the dual scaling expressions from Eq. (63) into the left and right sides respectively, we obtain

$$\begin{aligned} & \int_0^t \langle \bar{z}(\theta), \mathcal{S}_{21}\tilde{z}(t-\theta) + \mathcal{S}_{22}\tilde{w}(t-\theta) \rangle d\theta \\ &= \int_0^t \langle \bar{w}(\theta), \mathcal{S}_{11}\tilde{z}(t-\theta) + \mathcal{S}_{12}\tilde{w}(t-\theta) \rangle d\theta. \end{aligned}$$

Rearranging the terms to group by $\tilde{z}$ and $\tilde{w}$ yields

$$\begin{aligned} & \int_0^t \langle \mathcal{S}_{22}^* \bar{z}(\theta) - \mathcal{S}_{12}^* \bar{w}(\theta), \tilde{w}(t-\theta) \rangle d\theta \\ &= \int_0^t \langle -\mathcal{S}_{21}^* \bar{z}(\theta) + \mathcal{S}_{11}^* \bar{w}(\theta), \tilde{z}(t-\theta) \rangle d\theta. \end{aligned}$$

We observe that the terms inside the inner product match the definition of the dual scaled variables:

$$\tilde{\tilde{w}} = -\mathcal{S}_{21}^* \bar{z} + \mathcal{S}_{11}^* \bar{w}, \quad \tilde{\tilde{z}} = \mathcal{S}_{22}^* \bar{z} - \mathcal{S}_{12}^* \bar{w}.$$

Substituting these identities, we arrive at the scaled identity:

$$\int_0^t \langle \tilde{\tilde{z}}(\theta), \tilde{w}(t-\theta) \rangle d\theta = \int_0^t \langle \tilde{\tilde{w}}(\theta), \tilde{z}(t-\theta) \rangle d\theta. \quad (65)$$

This equation is structurally identical to (64), but relates the scaled primal and dual variables. Now, we proceed with the truncation argument. For any fixed $\tau > 0$, let $\tilde{w}(t) \in \mathbb{R}_{L_2}$ be arbitrary. Under the assumption that the primal and dual scaled system are well-posed, i.e. $\bar{\mathcal{S}}_{11} - \mathcal{D}^* \bar{\mathcal{S}}_{21}$ and $\mathcal{S}_{22}^* + \mathcal{D}^* \mathcal{S}_{21}^*$ are invertible, the scaled primal input can be defined as $\tilde{w}(t) = \tilde{\tilde{z}}(\tau - t)$ for $t \leq \tau$ and $\tilde{w}(t) = 0$ for $t > \tau$. Then

$$\begin{aligned} \|P_\tau \tilde{\tilde{z}}(t)\|_{\mathbb{R}_{L_2}}^2 &= \int_0^\tau \langle \tilde{\tilde{z}}(s), \tilde{w}(\tau - s) \rangle ds \\ &= \int_0^\tau \langle \tilde{\tilde{w}}(s), \tilde{z}(\tau - s) \rangle ds \\ &\leq \|P_\tau \tilde{\tilde{w}}(t)\|_{\mathbb{R}_{L_2}} \|P_\tau \tilde{z}(t)\|_{\mathbb{R}_{L_2}} \\ &\leq \|P_\tau \tilde{\tilde{w}}(t)\|_{\mathbb{R}_{L_2}} \|P_\tau \tilde{w}(t)\|_{\mathbb{R}_{L_2}}. \end{aligned}$$

Since by construction $\|P_\tau \tilde{w}(t)\|_{\mathbb{R}_{L_2}} = \|P_\tau \tilde{\tilde{z}}(t)\|_{\mathbb{R}_{L_2}}$, thus we conclude

$$\|P_\tau \tilde{\tilde{z}}(t)\|_{\mathbb{R}_{L_2}} \leq \|P_\tau \tilde{\tilde{w}}(t)\|_{\mathbb{R}_{L_2}}$$

Since $\mathcal{T}^{**} = \mathcal{T}$, Sublemma 1, and

$$\begin{bmatrix} (\mathcal{A}^*)^* & (\mathcal{B}^*)^* \\ (\mathcal{C}^*)^* & (\mathcal{D}^*)^* \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix}^* = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix}^{**} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix}$$

we have that sufficiency implies necessity.

B. Proof of Lemma 2

Given the operators $\mathcal{Q}, \mathcal{S}, \mathcal{R} \in \Pi_4$, write

$$\mathcal{V} := \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix}, \quad \mathcal{K} := \begin{bmatrix} -\mathcal{R} & \mathcal{S}^* \\ \mathcal{S} & -\mathcal{Q} \end{bmatrix}, \quad \bar{\mathcal{V}} = \mathcal{K}^{-1}.$$

Since $\mathcal{V}$ is a strict PN-PI operator, there exists $\epsilon > 0$ such that

$$\mathcal{Q} \succeq \epsilon I, \quad \mathcal{R} \preceq -\epsilon I.$$

so both $-\mathcal{R}$ and $\mathcal{Q}$ are self-adjoint, coercive, and invertible. Applying the block inverse formula to $\mathcal{K}$, generalized to the PI-operator setting from [5], gives the following expressions. In addition, [8, Theorem 2] shows that the inverse of a coercive, self-adjoint PI operator is again a coercive, self-adjoint PI operator.

$$\bar{\mathcal{Q}} = (\mathcal{S}^* \mathcal{Q}^{-1} \mathcal{S} - \mathcal{R})^{-1} \succeq \epsilon I,$$

and

$$\bar{\mathcal{R}} = -(\mathcal{Q} + \mathcal{S}(-\mathcal{R})^{-1} \mathcal{S}^*)^{-1} \preceq -\epsilon I.$$

Moreover, by composition algebra $\bar{\mathcal{S}}$ is also a PI operator,

$$\bar{\mathcal{S}} = \mathcal{Q}^{-1} \mathcal{S} \bar{\mathcal{Q}}.$$

Hence $\bar{\mathcal{V}}$ is a strict PN-PI operator. The converse follows by the same argument with $\mathcal{V}$ and $\bar{\mathcal{V}}$ interchanged, since

$$\mathcal{V} = \begin{bmatrix} -\bar{\mathcal{R}} & \bar{\mathcal{S}}^* \\ \bar{\mathcal{S}} & -\bar{\mathcal{Q}} \end{bmatrix}^{-1}.$$

Thus $\mathcal{V}$ is a strict PN-PI operator if and only if $\bar{\mathcal{V}}$ is a strict PN-PI operator.

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